

Teja Akella

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Education

Georgia Institute of Technology | GPA: 3.92

Atlanta, GA | Expected: December 2026

- Bachelor of Science in Computer Engineering w/ Cybersecurity, Systems and Architecture
 - Courses: Advanced Malware Analysis, GPU Programming, Operating Systems, Compilers and Interpreters, Networking, Computer Architecture, Malware Reverse Engineering, Programming HW/SW Systems
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Skills

Certifications: CompTIA Security+, CompTIA Network+, GFACT, and OSCP+ (In progress)

Technical Skills: OOP, Java, Python, JavaScript, TypeScript, React, C, Git, HTML, CSS, Go, Docker, SQL, x86, MIPS, MATLAB, AWS, GCP, TensorFlow, CUDA, RAG, TCP/UDP, OSI Model, Network Security, Web Security, Red Teaming, Wireshark, Nmap, Metasploit, BurpSuite, Kali, Ghidra, IDA Pro,

Work Experience

Security Engineering Intern | Capital One

McLean, VA | June 2026 - Present

- Developed **IAM infrastructure** and security automation tooling to streamline service lifecycle management, automate rehydration workflows, and improve operational resilience across enterprise environments.
- Conducted security, operational, and cost analyses of enterprise services during the Capital One–Discover merger, identifying **4 critical security gaps** and migration strategies expected to reduce annual operating costs by **>\$25K+** while strengthening service security.

AI Security Research Intern | Georgia Tech Cyber Forensics Innovation Lab

Atlanta, GA | March 2025 - Present

- Designed a novel detection framework for Architectural Backdoors and code-level triggers in state-of-the-art deep learning models by analyzing the model graph, activation patterns, and layer-level anomalies
- Investigated the security of federated learning systems under data poisoning attacks while designing a dynamic execution architecture for autonomous AI systems that improved the latency–accuracy Pareto frontier through adaptive model selection and orchestration, contributing to 3-4 research publications in AI security.

Security Penetration Tester | Automated Red Teaming

January 2025 - Present

- Performed white and black-box penetration testing on enterprise systems and open-source software, discovering critical vulnerabilities including RCE, XSS, and SSRF through manual exploitation and agentic red teaming.

Software Engineer Intern | NASA Ames Research Center

Mountain View, CA | June 2025 - August 2025

- Investigated **Graph-RAG** architectures for autonomous airspace safety systems, with findings contributing to a competitive research paper; engineered an application for NASA's HADES tool using **Electron.js, React, Redux, TypeScript**, and **PouchDB**, improving hazard identification speed by over **65%** through simulation-based analysis.

Founding Engineer | Startup | Reffy Inc.

Atlanta, GA | January 2024 - Present

- Designed and implemented a scalable, secure distributed infrastructure to reduce customer and enterprise labor by 80% through a client-facing web application and microservice backend. Currently partnered with many top universities, such as **Emory**. Expecting more than **50,000** users this year.

Research Intern | Purdue VIPER Lab

West Lafayette, IN | January 2024 - May 2024

- Conducted **deep learning** research on medical imaging, improving brain tumor detection accuracy by **30%** by designing and training custom lightweight CNN architectures.

Research Scholar | Air Force Research Lab

Dayton, OH | June 2022 - July 2023

- Supported Artemis 3 lunar mission prep by developing a moon surface simulation in **Unreal Engine 5** using the **JPL SPICE** protocol, enabling realistic astronaut training for South Pole operations.
 - Researched **mixed-reality** training methodologies, reducing RPG training costs by **85%** by prototyping a multiplayer simulator with **Distributed Interactive Simulation (DIS)** protocol and real-time **sensor tracking**.
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Projects

Malicious Network Traffic Detector & Visualization | Github/Takella6315/network-packet-tracing

- Developed a Python tool to parse PCAP files, map IPs to physical locations, and classify traffic as malicious using AI-based heuristics (DNS age, known C2 infrastructure, geolocation anomalies, and protocol behavior).
 - Technologies: Python, Wireshark, SMTP, Data Visualization, Network Protocol Analysis, Machine Learning, RAG
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Extra-Curricular Experiences:

GreyHat Cybersecurity Team | Vice-President

January 2025 - Present

- Compete in National CTF competitions (CSAW, CorCTF, KnightCTF, UCLACTF, BuckeyeCTF, etc.)
- Secured \$10K+ in sponsorship funding to support club growth and events